

## 6.13 Similarity in the Real World

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                             |                                                                                                              |                                                                                                      |                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark Focus     | MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.                                                                                                     |                                                                                                              |  Learning Target  | <ul style="list-style-type: none"><li>I can solve real-world problems involving similarity in two-dimensional figures.</li></ul> |
|  Benchmark Coherence | Building On<br>MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.                                                                                       |                                                                                                              | Working Toward<br>N/A                                                                                |                                                                                                                                  |
|  Essential Questions | <ul style="list-style-type: none"><li>How can similar triangles be used to solve real-world problems?</li><li>How can similar figures be used to plan large-scale structures?</li></ul>                                     |                                                                                                              |                                                                                                      |                                                                                                                                  |
|  Lesson Narrative    | This lesson is a culminating activity for the unit. Students will apply what they learned about similar figures to work through real-world scenarios to solve problems to find missing variables, angles, and side lengths. |                                                                                                              |                                                                                                      |                                                                                                                                  |
|  Component Summary  | 6.13.1 Warm-Up (~5 min.)                                                                                                                                                                                                    | Reflecting on the usefulness of math (SE p. 363)                                                             | 6.13.2 Activity (35-40 min.)                                                                         | Solving real-world problems using similar figures (SE p. 364)                                                                    |
|                                                                                                       | 6.13.3 Wrap-Up (~5 min.)                                                                                                                                                                                                    | Ticket Out the Door analyzing student work to determine a scale factor and find a segment length (SE p. 365) |                                                                                                      |                                                                                                                                  |
|  Resources         | Glossary                                                                                                                                                                                                                    |                                                                                                              | Materials                                                                                            | Additional Preparation                                                                                                           |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Similarity</li><li>Scale Factor</li><li>Proportional</li><li>Corresponding</li><li>Congruent</li></ul>                          |                                                                                                              | <ul style="list-style-type: none"><li>Growth Mindset Video</li><li>(Optional) Highlighters</li></ul> | <ul style="list-style-type: none"><li>6.13.1 Warm-Up contains a video to accompany the question prompts.</li></ul>               |

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| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Things to Consider                                                                                                                                                                                                                                                                                                                             |
|                                                                                                                | <ul style="list-style-type: none"><li>Students may only find the value of the variable without answering the question asked.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"><li>Consider these options for implementing the 6.13.2 Activity.<ul style="list-style-type: none"><li>Gallery Walk</li><li>Trivia game, such as Jeopardy-style or similar, where everybody can answer each question simultaneously</li><li>Task cards</li><li>Whiteboard group challenge</li></ul></li></ul> |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                             |
|                                                                                                                | <b>Take Turns</b><br>For 6.13.2 Activity, set students up to take turns in the work of the activity by having one person in the group talk through the problem while another group member writes. At each new problem, the two roles swap to other group members. Having one person talk while one person writes requires the talker to give clear and logical directions and requires the writer and other group members to listen carefully, check for understanding, and evaluate the accuracy of their peers' reasoning. | <b>MA.K12.MTR.7.1: Real-World Contexts</b><br>Students are expected to navigate a variety of real-world applications of similar figures for 6.13.2 Activity. Each of these questions allows students the opportunity to relate work done throughout this unit to the real world.                                                               |
| <br>Differentiate Instruction | Students will be exercising their comprehension of the topics by supplying background knowledge for 6.13.2 Activity. It would be valuable to prompt students to explore patterns that they may notice throughout each question when they have completed them. Students should be able to identify the critical features of each question and what steps are the most important to be able to solve each question.                                                                                                            |                                                                                                                                                                                                                                                                                                                                                |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                |

## 6.13.1 Warm-Up: With Math I CAN

**Component Overview:** Students watch a growth mindset video and then answer three questions independently.



### 6.13.1 Warm-Up: With Math I CAN

1. What struggles have you had in mathematics?  
*Answers will vary. In math, I have struggled with fractions and negatives.*
2. Students often ask teachers, “When will I ever use this?” What is one thing in mathematics you’ve wondered if you would ever use?  
*Answers will vary. I have wondered if I will ever use variables or proofs.*
3. Sometimes it isn’t that a job requires you to prove a theorem, but to use the logical thinking you learned in proving a theorem. Describe a few careers that you’re interested in and what can you do with math.  
*Answers will vary. I am interested in going into the medical field. I might use math there to measure and mix doses. I might go into automotive. I could use math there to calculate air and fuel ratios.*



Display the “With Math I Can” video for the entire class. After students watch this 1:41 minute video, prompt the students to read and answer the questions in 6.13.1 Warm-Up independently.

## 6.13.2 Activity: Real-World Problem Gallery Walk

**Component Overview:** Students answer each of six provided problems applying what they have learned in the unit to solve real-world problems.



### 6.13.2 Activity: Real-World Problem Gallery Walk

Complete each problem by recording your work and answer in the spaces provided.

| Problem   | Work                                                                                                                                                      | Answer                                           |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Problem A | $\frac{HR}{HM} = \frac{15}{6} = \frac{2}{5}$ $0.9 = \frac{2}{5}x$ $x = 2.25$                                                                              | The height of the tennis racket is 2.25 m.       |
| Problem B | $\frac{36}{12} = \frac{CP}{SC} = 3$ $3(4w - 5) = 3w + 3$ $w = 2$ $PN = 3 \cdot 2 + 3 = 9$                                                                 | The maximum depth for swimmers is 9 feet.        |
| Problem C | $\frac{855}{427.5} = 2$ $CM = 2 \cdot 527.5 = 1055$                                                                                                       | The lake is 1055 feet across.                    |
| Problem D | $\frac{74}{18.5} = 4$ $4(2x - 6.5) = 3x + 9$ $x = 7$ $\text{Perimeter} = PT + BT + HB + HP$ $\text{Perimeter} = 74 + 32 + (15 \cdot 4) + (3 \cdot 7 + 9)$ | The perimeter of quadrilateral HBTP is 196 feet. |
| Problem E | $0.5x - 5 = \frac{1}{562} \cdot (22x + 39)$ $x = 11$ $HA = 22 \cdot 11 + 39 = 281$ $\text{Perimeter} = 5 \cdot 281 = 1405$                                | The Pentagon has a perimeter of 1405 m.          |
| Problem F | $4(1.5y - 5) = 3y - 5$ $y = 5$ $HP = 3 \cdot 5 - 5 = 10$                                                                                                  | The length of $\overline{HP}$ is 10 ft.          |



Consider implementing one of the following for this activity.

- **Gallery Walk** – Post the problems around the room. Consider printing two sets of each problem to help spread students out. Students will work on one problem at a time before moving to the next problem. Students will record their work and answers in the table.
- **Task Cards** – Make cards of each problem that are then passed around the groups. Students will record their work and answers in the table.
- **Game** – Jeopardy or other game where each problem is displayed for all to see and work on simultaneously.
- **Whiteboard Group Challenge** – Divide the class into groups of three or four. Give each group member a different colored marker and a mini whiteboard. Display one problem at a time. Groups work together to answer the problem, but each member **MUST** work out the problem on their own whiteboard with their colored marker. When ready, call out for the students with a particular colored marker to stand up, showing their board. All groups whose member has the correct work and answer gets a point.



#### Common Misconception

Students may need a reminder not to just solve for variables but to also make sure they are fully answering each question completely.



Consider demonstrating how to use highlighters to organize information. Highlighters can be used to note important information in a problem or to highlight the corresponding angle or side length in the image one color. This provides a visual entry-point.

## 6.13.3 Wrap-Up: Ticket Out the Door

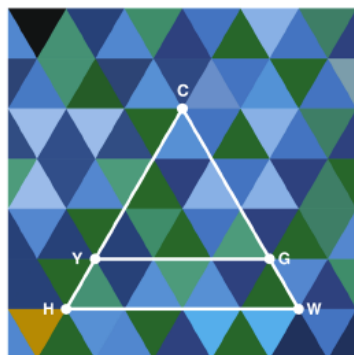
**Component Overview:** Students analyze two work samples to determine who correctly calculated the scale factor between two similar triangles. Students then solve for a side length using the scale factor.



### 6.13.3 Wrap-Up: Ticket Out the Door

1. Beth and Trevor are trying to find the lengths of the triangles on the quilt.

Beth and Trevor know that  $\triangle CYG \sim \triangle CHW$ ,  $CG = 21$  inches (in),  $YG = (2.5x + 1.5)$  in,  $GW = 7$  in, and  $HW = (6x - 14)$  in. Their work to find the scale factor is shown.



| Beth's Work                    | Trevor's Work                   |
|--------------------------------|---------------------------------|
| $\frac{CG}{GW} = \frac{21}{7}$ | $\frac{CG}{CW} = \frac{21}{28}$ |
| Scale Factor is 3              | Scale Factor is 0.75            |

- a. Whose work do you agree with?  
*I agree with Trevor.*
- b. What is the length of  $\overline{HW}$ ?  
 *$HW = 22$  in.*



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.



Remind students to check the validity of their answers by taking a step back and asking, "Does my answer make sense?" Example observation: If I am looking for a side length longer than the one given, my scale factor must be greater than 1.